

QUESTION-16

PRIORITY QUEUE

main.c	Output
<pre> 1 #include <stdio.h> 2 #define MAX 10 3 struct PriorityQueue {int data[MAX];int priority[MAX];int size; 4 }; 5 void insert(struct PriorityQueue *pq, int value, int pr) { 6 if (pq->size == MAX) { 7 printf("Queue Overflow\n"); 8 return; 9 } 10 pq->data[pq->size] = value; 11 pq->priority[pq->size] = pr; 12 pq->size++; 13 } 14 void deleteHighest(struct PriorityQueue *pq) { 15 if (pq->size == 0) { 16 printf("Queue Underflow\n"); 17 return; 18 } 19 int maxIndex = 0; 20 for (int i = 1; i < pq->size; i++) { 21 if (pq->priority[i] > pq </pre>	<pre> Element Priority 10 2 20 5 30 1 40 4 Deleted Element = 20 Element Priority 10 2 30 1 40 4 === Code Execution Successful === </pre>

main.c	Output
<pre> 22 ->priority[maxIndex]) 23 maxIndex = i; 24 } 25 printf("Deleted Element = %d\n", pq ->data[maxIndex]); 26 for (int i = maxIndex; i < pq->size - 1; i ++){ 27 pq->data[i] = pq->data[i + 1]; 28 pq->priority[i] = pq->priority[i + 1]; 29 } 30 pq->size--; 31 void display(struct PriorityQueue pq) { 32 if (pq.size == 0) { 33 printf("Queue is Empty\n"); 34 return; 35 } 36 printf("\nElement\tPriority\n"); 37 for (int i = 0; i < pq.size; i++) { 38 printf("%d\t%d\n", pq.data[i], pq .priority[i]); 39 } 40 } </pre>	<pre> Element Priority 10 2 20 5 30 1 40 4 Deleted Element = 20 Element Priority 10 2 30 1 40 4 === Code Execution Successful === </pre>

```

41 int main() {
42     struct PriorityQueue pq;
43     pq.size = 0;insert(&pq, 10, 2);insert(&pq,
        20, 5);insert(&pq, 30, 1);insert(&pq,
        40, 4);
44     display(pq);
45     deleteHighest(&pq);
46     display(pq);
47     return 0;
48 }

```